# A note on Open AI's "A counterexample to Hadwiger's conjecture"

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## 1 Summary

The manuscript constructs a triangle-free graph $G$ whose complement $H = \bar{G}$ satisfies $\alpha(H) \leq 2$ and $\mathrm{cm}(H) < m/100$, where $m$ is the number of vertices. The connected matching number is defined as follows:

**Definition 1.** *A matching $M$ in a graph $H$ is called a* connected matching *if every two distinct edges of $M$ are joined by an edge of $H$: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \leq h(H)$, where $h(H)$ is the largest integer $t$ such that $H$ contains a $K_t$ minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \leq \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \leq 2$ so that every color class has at most two vertices, we obtain, for $m$ large enough,

$$h(\bar{G}) \leq \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \leq \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

## 2 The construction

### 2.1 Triangle-free graphs from linear algebra

We first describe the general linear-algebraic scheme and then specify the data used in the manuscript. All vector spaces, ranks, and matrix operations are over $\mathbb{F}_2$.

Let $V, W$ be finite-dimensional vector spaces. Fix $a \in V^*$ and a symmetric bilinear form $T : V \times V \to \mathbb{F}_2$, and put

$$r(\lambda) = a + T(\lambda, \cdot) \in V^*.$$

Let $\Omega$ be the set of vertices. For each $o \in \Omega$, assign an injective linear map $U_o : V \to W$ and a linear functional $u_o \in W^*$. Then we can define the associated graph:

**Definition 2.** *Two vertices $i, j \in \Omega$ are adjacent if there exist $\lambda_i, \lambda_j \in V$ such that*

$$U_i\lambda_i = U_j\lambda_j, \tag{1}$$

$$u_j \circ U_i = r(\lambda_i), \qquad u_i \circ U_j = r(\lambda_j), \tag{2}$$

$$a(\lambda_i) + a(\lambda_j) = 1. \tag{3}$$

*Equation* (2) *is an equality in $V^*$.*

**Lemma 1.** *The associated graph is loopless and triangle-free.*

*Proof.* Loopless is guaranteed by injectivity of $U$ and (3). Suppose $i, j, k$ is a triangle. Write $\lambda_{ij}$ for the $\lambda_i$ at $i$ for the pair $ij$. Evaluate (2) on $\lambda_{ik}$:

$$u_j(U_i\lambda_{ik}) = a(\lambda_{ik}) + T(\lambda_{ij}, \lambda_{ik}).$$

Then sum six equations by permuting $i, j, k$. For fixed $j$, the two terms on the left cancel because $U_i\lambda_{ik} = U_k\lambda_{ki}$. For fixed $i$, the two bilinear terms cancel by symmetry of $T$. The remaining right-hand side is

$$\sum_{\{i,j\}} \big(a(\lambda_{ij}) + a(\lambda_{ji})\big) = 1 + 1 + 1 = 1,$$

whereas the left-hand side is zero, a contradiction. □

## 2.2 Construction of $U$ and $u$

Write $\mathrm{Mat}_{m,n}(\mathbb{F}_2)$ for the space of $m \times n$ binary matrices, and $\mathrm{Mat}_n(\mathbb{F}_2) = \mathrm{Mat}_{n,n}(\mathbb{F}_2)$. Fix a finite index set $\mathcal{E}$, positive integers $N, d_B, h$ with $N \geq 2(d_B + h)$, and a matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$, to be specified later. Take

$$V \subseteq \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_{d_B}(\mathbb{F}_2) \quad \text{a linear subspace,} \qquad W = \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_N(\mathbb{F}_2).$$

For $x \in V$, write $x = (x_e)_{e \in \mathcal{E}}$, so $x_e \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$.

At each component $e$, consider quadruples

$$P_e, Q_e \in \mathrm{Mat}_{N,d_B}(\mathbb{F}_2), \qquad X_e, Y_e \in \mathrm{Mat}_{N,h}(\mathbb{F}_2)$$

satisfying

$$\begin{aligned}&[P_e\ X_e] \text{ and } [Q_e\ Y_e] \text{ have full column rank,}\\ &P_e^T Q_e = E,\\ &X_e^T Q_e = 0, \qquad P_e^T Y_e = 0.\end{aligned}$$

The condition $N \geq 2(d_B + h)$ ensures there is enough admissible quadruples. Let $\mathcal{O}$ be the set of tuples $o = (P_{o,e}, Q_{o,e}, X_{o,e}, Y_{o,e})_{e\in\mathcal{E}}$ of admissible quadruples. Define a probability law on $\mathcal{O}$ by sampling uniformly at each component, independently across components.

For matrices of the same size, define $\langle A, B\rangle = \operatorname{Tr}(A^T B)$. Each $o \in \mathcal{O}$ determines $U_o : V \to W$ and $u_o \in W^*$ by

$$U_o(x) = (P_{o,e} x_e Q_{o,e}^T)_{e\in\mathcal{E}}, \qquad u_o(\xi) = \sum_{e\in\mathcal{E}} \langle Y_{o,e} X_{o,e}^T, \xi_e\rangle.$$

for $x \in V$ and $\xi \in W$. Each image component $P_{o,e} x_e Q_{o,e}^T$ belongs to $\mathrm{Mat}_N(\mathbb{F}_2)$. Left inverses of $P_{o,e}$ and $Q_{o,e}$ recover $x_e$ from that component, so $U_o$ is injective. Thus these maps provide the vertex data required in Section 2.1; the particular subspace $V$ and the forms $a, T$ will be specified below.

Moreover,

$$u_o(U_o(x)) = \sum_{e\in\mathcal{E}} \operatorname{Tr}\big(X_{o,e} Y_{o,e}^T P_{o,e} x_e Q_{o,e}^T\big) = 0,$$

since $Y_{o,e}^T P_{o,e} = 0$.

### 2.3 Parameters

The only asymptotic parameter is $n$. The manuscript fixes

$$C_0 = 1000, \qquad g = 10^9 + 1, \qquad D = 4C_0 g, \qquad M = 2^{1000}.$$

The constants $b, j_*, J, r_0, M_0$ are chosen in the dependency order of Appendix A of the manuscript, before $n$ tends to infinity. Their numerical values are not specified. Set

$$h = 1000 r_0, \qquad N = M_0 n, \qquad m = 2^{C_0 g N}.$$

Here $M$ is a density bound used in the distribution theorem, whereas $M_0$ scales the ambient dimension. Choose $M_0$ sufficiently large to satisfy the dimension and counting requirements, including $N \geq 2(d_B + h)$ for all sufficiently large $n$. We take $n \geq 2r_0$ and $j_* \geq 1$.

### 2.4 Boolean moment spaces

Let

$$I = \mathbb{Z}/g\mathbb{Z}, \quad \mathcal{E} = \binom{I}{2}, \quad I_d = \{d+1, \ldots, d+(g-1)/2\} \subset I.$$

Elements of $I$ are *tags* and elements of $\mathcal{E}$ are *components*. A *Boolean point* consists of a *selector* $s \in \mathbb{F}_2^b$ and *base coordinates* $z$ in $g^2+3$ blocks of $n$ bits:

$$z = \left((z_{O_{d,t}})_{d,t\in I},\ z_S,\ z_\#,\ z_Z\right) \in (\mathbb{F}_2^n)^{g^2+3}.$$

For each block label $A \in \{O_{d,t} : d, t \in I\} \cup \{S, \#, Z\}$, write

$$z_A = (z_{A,1}, \ldots, z_{A,n}) \in \mathbb{F}_2^n.$$

Thus $z$ has $g^2+3$ blocks, each containing $n$ bits; $z_{A,k}$ denotes the $k$th bit in block $A$. Write

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \qquad B = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}, \qquad d_B = \dim B.$$

Let $p(s) \in \mathbb{F}_2^{p_*}$ be the bit vector listing the evaluations at $s$ of all squarefree selector monomials of degree at most $j_*$, in a fixed order. Define the evaluation vector and its point moment by

$$v(s,z) = p(s) \otimes (1,z), \qquad w(s,z) = v(s,z)v(s,z)^T \in \mathrm{Mat}_{d_B}(\mathbb{F}_2).$$

Identify $B \otimes B$ with $\mathrm{Mat}_{d_B}(\mathbb{F}_2)$ using the fixed coefficient basis. Let $\mathcal{M} \subseteq \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ be the span of all point moments. For $\ell \in I$, let $\mathcal{M}_\ell$ be the span of those point moments whose base values satisfy $O_{d,t} = 0$ whenever $\ell \notin I_d$. Let $\mathcal{M}_*$ be the span obtained by setting all $O$-blocks to zero.

Define the linear cut-profile map

$$\begin{aligned} f : \bigoplus_{\ell\in I} \mathcal{M}_\ell &\longrightarrow \bigoplus_{e\in\mathcal{E}} \mathcal{M}, \\ (w_\ell)_{\ell\in I} &\longmapsto \left(w_\ell + w_{\ell'}\right)_{\{\ell,\ell'\}\in\mathcal{E}}. \end{aligned}$$

Take the witness space in Sections 2.1 and 2.2 to be $V = \operatorname{im} f$. A tuple $(w_\ell)_{\ell\in I}$ is a representation of $x \in V$ if $f((w_\ell)_{\ell\in I}) = x$. By Lemma 2.3 of the manuscript,

$$\ker f = \{(w_*)_{\ell\in I} : w_* \in \mathcal{M}_*\},$$

that is, two representations of the same $x$ differ by a constant tuple with value in $\mathcal{M}_*$.

### 2.5 The functional $a$, the bilinear form $T$, and the matrix $E$

The witness space is now fixed as $V = \operatorname{im} f$, and the ambient space remains $W = \bigoplus_{e\in\mathcal{E}} \mathrm{Mat}_N(\mathbb{F}_2)$ from Section 2.2. We next specify $a \in V^*$, a symmetric bilinear form $T$ on $V$, and the matrix $E$ used to constrain the frames.

Define *linear testers* on moments by selecting ordered matrix entries:

$$\eta_{d,t}(w(s,z)) = \sum_{\rho=1}^{r_0} z_{O_{d,t},2\rho-1} z_{O_{d,t},2\rho},\ d,t \in I$$

$$\eta_S(w(s,z)) = \sum_{\rho=1}^{r_0} z_{S,2\rho-1} z_{S,2\rho}.$$

Put $\eta_O = \sum_{d,t} \eta_{d,t}$ and $\eta = \eta_O + \eta_S$. These are elements of $\mathcal{M}^*$ and take values in $\mathbb{F}_2$.

For a cut profile $x$ represented by $(w_\ell)$, set

$$a_t(x) = \sum_{\ell,d} \eta_{d,t}(w_\ell), \qquad a(x) = \sum_t a_t(x), \qquad b_t(x) = \sum_{\ell\neq t} \eta_S(w_\ell).$$

These functionals do not depend on the representation: the $O$-testers vanish on $\mathcal{M}_*$, and a constant change contributes $(g-1)\eta_S(w_*) = 0$ to $b_t$.

Define

$$T_0(x,y) = a(x)a(y) + \sum_{t\in I} (a_t(x)b_t(y) + b_t(x)a_t(y)).$$

For each $1 \le j \le J$ and ordered pair $e, f \in \mathcal{E}$, fix endomorphisms $L_j^{ef}, R_j^{ef}$ of $B$, chosen to satisfy the uniform mixing conditions of Lemma 5.5 of the manuscript. With the entrywise pairing $\langle A, B\rangle = \mathrm{Tr}(A^T B)$, put

$$B_1(x,y) = \sum_{j=1}^{J} \sum_{e,f\in\mathcal{E}} \left\langle x_e, L_j^{ef} y_f (R_j^{ef})^T \right\rangle,$$

$$T_1(x,y) = B_1(x,y) + B_1(y,x), \qquad T = T_0 + T_1.$$

Then $T$ is symmetric and $T(x,x) = a(x)$. This specifies $r(\lambda) = a + T(\lambda,\cdot)$ in the edge definition.

The matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ prescribes $P_{o,e}^T Q_{o,e} = E$ in Section 2.2. Let $E_\eta$ be the sum of the ordered tester matrices, so $\langle E_\eta, w\rangle = \eta(w)$ for $w \in \mathcal{M}$. Choose $M_1, \dots, M_b \in \mathrm{Mat}_n(\mathbb{F}_2)$ along with the mixers as in Lemma 5.5, and define $E_\#$ by

$$v(s,z)^T E_\# v(s',z') = \sum_{\rho=1}^{b} (s_\rho + s'_\rho) z_\#^T M_\rho z'_\#.$$

This is bilinear on $B$ since $j_* \ge 1$. Set $E = E_\eta + E_\#$. Its additional term vanishes on coincident point inputs, so

$$\langle E, w\rangle = \eta(w) \qquad (w \in \mathcal{M}).$$

The $Z$-block does not occur in $E$. The mixers and the matrices $M_\rho$ are chosen once for each sufficiently large $n$, before graph sampling. Their existence with the required rank properties is supplied by Lemma 5.5; arbitrary choices need not give the connected-matching bound.

### 2.6 The vertex law

With $\mathcal{E}, d_B, N, h, V, a, T, E$ now specified, apply Section 2.2. Let $\mathcal{O}_n$ be the set of admissible frame tuples and let $\mu_n$ be its product uniform law across components. An element $o \in \mathcal{O}_n$ determines the maps $U_o : V \to W$ and $u_o \in W^*$ already defined there. In the general scheme of Section 2.1, take $\Omega = \mathcal{O}_n$.

The initial random variable is therefore a tuple of frame matrices. The coefficient space, moment spaces, $a, T$, and $E$ are fixed before drawing these tuples. In particular, the matrix $E$ alone does not specify the graph: adjacency also depends on $V, a, T$.

### 2.7 The sampled graph and the remaining theorem

Sample

$$o_1, \dots, o_m \overset{\text{iid}}{\sim} \mu_n, \qquad m = 2^{C_0 gN}.$$

On the positions $1, \dots, m$, put $ij \in E(G)$ exactly when $o_i, o_j$ satisfy the edge criterion (1)–(3); put $ij \in E(H)$ otherwise. This completes the definition of the random graph. Its edges are determined by the sampled frames, with no further independent edge sampling.

**Remark 1.** *The initial random variable is the tuple of frame matrices $o$, not a Boolean point $(s, z)$. The monomial list and moment spaces are fixed globally. Boolean points vary when constructing witnesses; sampling point parameters with the frames fixed is called a query in the manuscript.*

Triangle-freeness of $G$, and hence $\alpha(H) \le 2$, holds for every sample. The harder assertion is that $\mathrm{cm}(H) < m/100$ with probability $1 - \exp(-\Omega(m))$ (Proposition 3.4). To obtain it, the manuscript calls an ordered pair of raw vertices that do not satisfy the edge criterion a *unit*; such a pair represents an edge of $H$ when placed at distinct sampled positions. Two units conflict if all four cross pairs are edges of $G$, so no edge of $H$ joins their endpoint sets. Their corresponding edges therefore cannot coexist in a connected matching.

Theorem 3.1 asserts that every law $\sigma$ on units satisfying the pointwise density bounds

$$\sigma_1 \le M\mu_n, \qquad \sigma_2 \le M\mu_n, \qquad \sigma \le 2^{DN}(\mu_n \otimes \mu_n)$$

gives a conflict between two independent $\sigma$-units with probability at least $2^{-100gN}$. The endpoints within one unit may be dependent. An entropy-based container argument transfers this statement to the sampled graph. Thus the construction above specifies the graph; the subsequent distribution theorem and container argument establish the small connected matching number needed for the counterexample.

# A The clique-minor bound (Proposition 3.5)

**Proposition 1.** *For every finite nonempty graph $H$ with $m$ vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

*Proof.* Write $c = \mathrm{cm}(H)$ and $b = h(H)$. A $K_b$ minor model consists of $b$ disjoint nonempty connected branch sets, with an edge joining every pair. Let $s$ be the number of singleton branch sets and $e$ the number of two-vertex branch sets.

Each two-vertex branch set supplies an edge. The singleton vertices form a clique, so pairing them supplies $\lfloor s/2 \rfloor$ more edges. Together these form a connected matching, since every pair of its edges is joined by an edge of $H$. Thus

$$c \ge e + \left\lfloor \frac{s}{2} \right\rfloor, \qquad s \le 2(c - e) + 1.$$

Every remaining branch set has at least three vertices, so

$$m \ge s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Consequently,

$$3b \le m + 2s + e \le m + 4c - 3e + 2 \le m + 4c + 2,$$

which proves the claim. □